

I've been thinking about the K4 cipher, and I wanted to share some of my observations.

After some analysis, I think that JS chose the crib words "BERLIN" and "CLOCK" with the belief that they would be enough information to break the cipher. When that clue failed, he added "EASTNORTHEAST" to the mix.

As a result, we have two sizable cribs, each solidly within their half of the cipher:

```
-----EASTNORTHEAST-----  
-----BERLINCLOCK-----
```

However, there isn't much overlap between these halves when folded like this, especially since "BERLIN" and "NORTHEAST" don't correspond to anything in the other half.

But something interesting happens when you reverse one of the halves. Now the two cribs have significant positional overlaps! Maybe this is why "NORTHEAST" was the next clue after "BERLINCLOCK"?

```
                EASTNORTHEAST  
?OBKRUXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJSSEKZZ  
                KCOLCNILREB  
RACKEUAUHUHGIDCJTXZKDGWKPFZMTTPYBNBFNIWAIDULKJTAW
```

Because the cipher has an odd length of 97 characters (i.e., ignoring the question mark), I don't think JS intended for the halves to be directly compared to one another. Rather, I've been working on ideas that involve reversing either the plaintext or ciphertext (it's been easier to reverse the cipher so that the plaintext remains readable):

```
-----EASTNORTHEAST-----BERLINCLOCK-----  
OBKRUXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJSSEKZZWATJKLUDIAWINFBNYPVTMTZFPKWGDKZXTJCDIGKUHUAUEKCAR  
RACKEUAUHUHGIDCJTXZKDGWKPFZMTTPYBNBFNIWAIDULKJTAWZZKESQJSQWTWOSKGNRPQQVRLFWBBFILOSBLUHGOXOURKBO  
-----KCOLCNILREB-----TSAEHTRONTSAE-----
```

I've also been thinking about how JS might have generated the ciphertext given the plaintext (or reverse text). Maybe he used an autokey? Unfortunately, I couldn't make it work, but I did find one particularly interesting coincidence that makes me think this line of thinking has at least a little bit of merit.

Consider this portion with the reversed ciphertext:

BERLINCLOCK (plaintext)
SSKGNRPQQVR (ciphertext, reversed)

I tried to figure out what key would work:

B:? = S
E:? = S
R:? = K
L:? = G
I:? = N
N:? = R

If we use the full tableau from the sculpture, then the regular English alphabet maps into the modified KRYPTOS alphabet, so A:A=K, B:A=R, C:A=Y, etc. (I can't shake the feeling that the dual alphabets are important for K4 whereas the KRYPTOS alphabet alone was used for K1-K2)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V
U	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X
W	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y

To find the missing keytext, locate the plaintext letter across the top, follow that column down until you arrive at the ciphertext, and then follow that row to the left to get the keytext.

B:F=S
E:C=S
R:J=K
L:C=G
I:L=N
N:

	A	B	C	D	E	F	G
A	K	R	Y	P	T	O	S
B	R	Y	P	T	O	S	A
C	Y	P	T	O	S	A	B
D	P	T	O	S	A	B	C
E	T	O	S	A	B	C	D
F	O	S	A	B	C	D	E
G	S	A	B	C	D	E	F

BERLINCLOCK (plaintext)
SSKGNRPQQVR (ciphertext)
FCJCLOBJGUR (keytext)

What piqued my interest here is that CLOBJ portion... which is really close to CLOCK (if you shift the B and J by +1). And this is only offset by three characters from the plaintext, so maybe this is a form of autokey? The starting key could then be something like the YAR offset letters (which is RAY... reversed!).

If this is a modified autokey, the next three characters after CLOCK are GUR... If you happen to shift each of these +1, you get HVS, which could be read as "HAS".

FCJCLOBJGUR (autokey)
66400011111 (shift)
LINCLOCKHVS (corresponding plaintext)

At this point, I couldn't derive a repeating pattern needed to shift the autokey for this to work universally. And, there seems to be nothing but noise on the EASTNORTHEAST side of the cipher when applying this method. So, I think I'm at a dead end here trying to make an autokey based on reversing text work.

But, maybe somebody else will see a pattern that I can't?